

Mini-Project Proposal: 6-bit, 1 MS/s SAR ADC

Advanced Microelectronics with Cadence — JUNIA / IEMN

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1. Objective

We will design and simulate a 6-bit successive-approximation-register (SAR) analog-to-digital converter. Following the course flow, we first define the specifications and architecture, then build a behavioral model and testbench, and finally replace blocks with transistor-level (analog) and RTL/gate-level (digital) implementations for mixed-signal simulation. The SAR architecture is a good fit because it combines a small analog core (comparator, capacitor array) with a digital control FSM, which exercises both halves of the course.

2. Target Specifications

Parameter	Value	Parameter	Value
Resolution	6 bits	Sampling rate	1 MS/s ($T_s = 1\ \mu\text{s}$)
Reference voltage	$V_{ref} = 1\ \text{V}$	Input range	0–1 V (single-ended)

3. Proposed Architecture

The input is sampled onto a binary-weighted capacitive DAC (CDAC). On each clock cycle a StrongARM comparator compares the sampled input voltage with the trial voltage generated by the CDAC using the reference voltage, and the SAR logic keeps or drops the bit under test, performing a binary search from MSB to LSB. With 1 sampling cycle plus 6 comparison cycles, an initial internal clock target of approximately 8 MHz (125 ns period) is proposed, providing timing margin for CDAC settling and comparator operation within the $1\ \mu\text{s}$ conversion period.

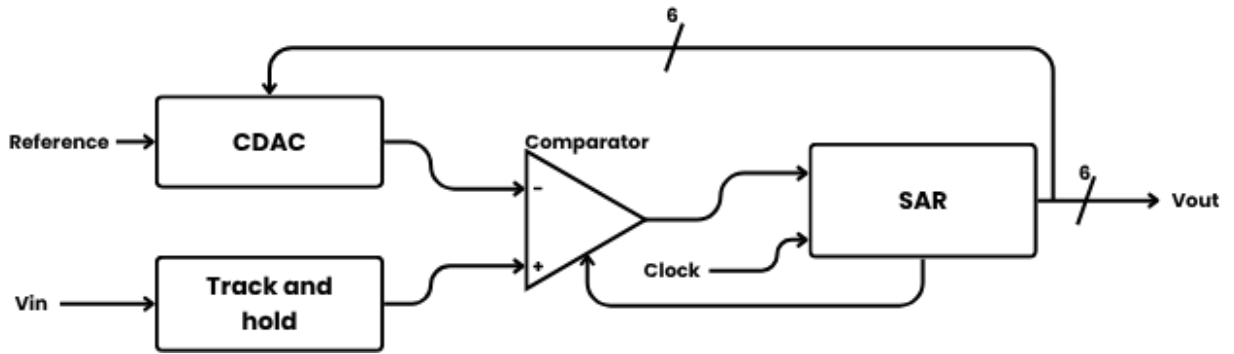


Figure 1: Block Diagram of the 6-bit SAR ADC

Blocks:

- **CDAC:** binary-weighted array ($32C, 16C, 8C, 4C, 2C, C$ plus a dummy C , $64 C_u$ total); unit capacitor sized for matching and kT/C noise well below $\frac{1}{2}$ LSB.
- **Comparator:** StrongARM latch (dynamic, clocked, low static power), target input-referred offset: $< \frac{1}{2}$ LSB.
- **Track & Hold:** MOS switch sampling V_{in} onto the CDAC top plate (charge injection and settling to be checked).
- **SAR logic:** synchronous FSM written in HDL (SystemVerilog/Verilog), synthesized to gates.
- **Clock:** external ideal source, with non-overlapping phases for sample and convert.

4. Work Plan and Deliverables

1. **Specifications and architecture** (this proposal).
2. **Behavioral model** of the full ADC (Verilog-A / Verilog-AMS) and testbench: ramp and sine inputs.
3. **Block-level design and simulation** (at least one block, as required): StrongARM comparator and CDAC in Virtuoso (transient, offset, and speed); SAR FSM in RTL with Xcelium simulation.
4. **Mixed-signal integration:** replace behavioral blocks with the transistor-level and gate-level netlists.
5. **Performance extraction** (optional): offset, gain error, DNL, INL, SNR/ENOB from a coherent-sine or ramp test.
6. **Final presentation** (W46) and report.

Success criteria: Success criteria: correct 6-bit output codes across the 0–1 V input range at 1 MS/s, with the ADC meeting the specified functional and timing requirements.